Replication: LinkedIn(to) Job Opportunities

Experimental Evidence from Job Readiness Training (Wheeler, Garlick, Johnson, Shaw, and Gargano, AEJ: Applied 2022)

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Contents

1 The paper in one page

The question is whether teaching work seekers to use LinkedIn raises their employment, and the answer is a 7 percentage point gain that persists for a year. Harambee, a South African job-readiness program, ran 30 training cohorts in 2016. Cohorts were paired into 14 stratification blocks by city and date, and one cohort per block was randomized to receive a LinkedIn training module inside the program. Treated candidates were 31 percentage points more likely to hold a LinkedIn account and 7 percentage points more likely to be employed at the end of the program, with effects of the same size 6 and 12 months later.

The identification is the randomization; the mechanism is the harder question and the paper treats it as such. The experiment identifies the effect of the training on employment and the effect on every LinkedIn measure (Table 2). It does not identify how much of the employment gain runs through LinkedIn use, because nobody randomized LinkedIn use. The paper assembles four pieces of evidence in Section IV: the nonexperimental association between account ownership and employment (Table 4), effect heterogeneity by baseline account (Table 5), effects on candidate mechanisms (Table 6), and, in Appendix C, two "share explained" statistics under stated assumptions. It says that none of this is prespecified and that the assumptions are strong.

Why this paper anchors Section 10. It is a clean experiment whose authors reach for mediation only with explicit caveats. The Appendix C shares are the linear mediation model of the lecture, estimated as a system with delta-method standard errors, and the mechanism tables are the "effects on mediators" step that every mediation analysis should report first. The public data reproduce all of it.

What the public replication package holds. `panel_clean_final.dta` (1,638 candidates, one row each, with the three survey waves stacked as variables), the Stata scripts for the paper and the appendices, the auxiliary programs (wild bootstrap, extensive/intensive decomposition), and the `.tex` and `.csv` artifacts of every table. The LinkedIn usage components behind the intensive-margin indices were redacted by LinkedIn before release, so four columns of Table 6 and the component detail of Table 2 cannot be rebuilt. Every reproduced number below is printed next to the published one.

2 Data and specifications

```r
wh <- as.data.frame(haven::zap_labels(haven::read_dta(file.path(data_path, "panel_clean_final.dta"))))
wh$strata <- factor(wh$strata)
wh$cohort <- factor(wh$studref)
c(candidates = nrow(wh), treated = sum(wh$treated), cohorts = nlevels(wh$cohort), blocks = nlevels(wh$strata))
#> candidates    treated    cohorts     blocks
#>       1638        890         30         14
```

Every treatment effect in the paper is one regression. The outcome is regressed on the treatment indicator with stratification-block fixed effects, and standard errors are clustered by cohort (30 clusters). In Stata this is `reghdfe y treated, absorb(strata) cluster(studref)`; here it is one `fixest::feols` call. The helper below returns the coefficient, the standard error, the control-group mean, and the sample sizes the paper's tables report. `fixest`'s default small-sample correction matches `reghdfe`'s for this design (block fixed effects are not nested within cohorts), which the tables confirm to three decimals.

```r
itt <- function(y, data = wh, cond = NULL) {
  d <- data[!is.na(data[[y]]), ]
  fit <- feols(as.formula(paste(y, "~ treated | strata")), data = d, cluster = ~studref)
  ct <- coeftable(fit)
  out <- tibble(outcome = y, estimate = ct["treated", "Estimate"], std.error = ct["treated", "Std. Error"],
                control_mean = mean(d[[y]][d$treated == 0]), n = nobs(fit), n_cohorts = length(unique(d$studref)),
                adj_r2 = unname(r2(fit, "ar2")))
  if (!is.null(cond)) out$control_mean_cond <- mean(d[[y]][d$treated == 0 & d[[cond]] == 1], na.rm = TRUE)
  out
}
```

Covariates enter only in Table 4 and in our additions, with the paper's missing-value convention. The do-file replaces missing education dummies by zero and, for every other covariate with missing values, adds a missing indicator and a zero-filled copy. We build the same list.

```r
covs <- c("age", "gender_composite", "prev_employ", "cog_num", "cog_comm", "cog_cft",
          "edn_dummy3", "edn_dummy4", "region1", "region2")
covs_miss <- character(0)
for (v in covs) {
  if (v %in% c("edn_dummy3", "edn_dummy4")) {
    wh[[v]][is.na(wh[[v]])] <- 0
```

```r
    covs_miss <- c(covs_miss, v)
  } else if (anyNA(wh[[v]])) {
    wh[[paste0(v, "_miss")]] <- as.integer(is.na(wh[[v]]))
    wh[[paste0(v, "_nomiss")]] <- ifelse(is.na(wh[[v]]), 0, wh[[v]])
    covs_miss <- c(covs_miss, paste0(v, "_miss"), paste0(v, "_nomiss"))
  } else {
    covs_miss <- c(covs_miss, v)
  }
}
covs_miss
#>  [1] "age_miss"                "age_nomiss"
#>  [3] "gender_composite_miss"   "gender_composite_nomiss"
#>  [5] "prev_employ_miss"        "prev_employ_nomiss"
#>  [7] "cog_num_miss"            "cog_num_nomiss"
#>  [9] "cog_comm_miss"           "cog_comm_nomiss"
#> [11] "cog_cft_miss"            "cog_cft_nomiss"
#> [13] "edn_dummy3"              "edn_dummy4"
#> [15] "region1"                 "region2"
```

3 Table 1: sample characteristics and balance

Table 1 establishes that randomization produced comparable arms, and the paper's balance test is a regression. For each characteristic the do-file regresses the variable on the two arm indicators without a constant, clustered by cohort, and tests their equality; that is the clustered t-test of the treatment coefficient. The standardized difference divides the arm gap by the pooled standard deviation. Two characteristics are cohort-level (size and completion rate), and their tests run on 30 cohort observations.

```r
table1_row <- function(v, label, binary = FALSE, cohort_level = FALSE) {
  if (cohort_level) {
    d <- wh %>% group_by(studref) %>% summarise(x = first(.data[[v]]), treated = first(treated), .groups = "drop")
    fit <- lm(x ~ treated, data = d)
    p <- coeftable(feols(x ~ treated, data = d, vcov = "hetero"))["treated", "Pr(>|t|)"]
  } else {
    d <- wh[!is.na(wh[[v]]), ]; d$x <- d[[v]]
    p <- coeftable(feols(x ~ treated, data = d, cluster = ~studref))["treated", "Pr(>|t|)"]
  }
  q <- quantile(d$x, c(0.1, 0.9))
  tibble(Variable = label, N = nrow(d), Mean = mean(d$x), `Std. dev.` = sd(d$x),
         `10th` = if (binary) NA else q[[1]], `90th` = if (binary) NA else q[[2]],
         `p-value` = p, `Std. diff.` = (mean(d$x[d$treated == 1]) - mean(d$x[d$treated == 0])) / sd(d$x))
}
t1 <- bind_rows(
  table1_row("age", "Age"), table1_row("cog_num", "Numeracy score"), table1_row("cog_comm", "Communications score"),
  table1_row("cog_cft", "Cognitive score"), table1_row("gender_composite", "Gender", binary = TRUE),
  table1_row("edn_over12", "High school education", binary = TRUE),
  table1_row("edn_postsecond", "Post-secondary education", binary = TRUE),
  table1_row("edn_uni", "University education", binary = TRUE), table1_row("prev_employ", "Previously employed", binary = TRUE),
  table1_row("cohort_size", "Size of cohort", cohort_level = TRUE),
  table1_row("bridge_p_complete", "Program completion rate", cohort_level = TRUE))
published1 <- read.csv(file.path(data_path, "output", "summstats.csv"), skip = 1, header = TRUE, check.names = FALSE)
t1$`Published p` <- published1[["p-value"]][match(t1$Variable, trimws(published1$Variable))]
t1$`Published std. diff.` <- published1[["Std Diff"]][match(t1$Variable, trimws(published1$Variable))]
t1 %>% mutate(across(where(is.numeric), ~ round(.x, 2))) %>%
  kbl(booktabs = TRUE, linesep = "", digits = 2, caption = "Table 1 of the paper: sample characteristics, balance p-values, and
  tab_style("scale_down")
```

The reproduced p-values and standardized differences match the published ones, including the one communication-score imbalance at p = 0.05. The paper reads that imbalance as chance among eleven tests and returns to it in Appendix B7 with heterogeneity by communication skill. Balance on baseline employment (previously employed, 38 percent) matters most for what follows, because previous employment is the strongest predictor of both LinkedIn use and later employment (Table 4).

Table 1: Table 1 of the paper: sample characteristics, balance p-values, and standardized differences, with the published p-values and standardized differences in the last two columns.

Variable	N	Mean	Std. dev.	10th	90th	p-value	Std. diff.	Published p	Published std. diff.
Age	1636	23.65	2.95	19.89	27.66	0.11	-0.16	0.11	-0.16
Numeracy score	1547	-0.03	1.00	-1.48	1.32	0.59	-0.06	0.59	-0.06
Communications score	1610	0.08	0.96	-1.03	1.18	0.05	0.15	0.05	0.15
Cognitive score	1617	0.04	0.98	-1.32	1.66	0.52	0.07	0.52	0.07
Gender	1633	0.61	0.49	NA	NA	0.49	-0.05	0.49	-0.05
High school education	1500	0.99	0.08	NA	NA	0.39	0.06	0.39	0.06
Post-secondary education	1500	0.38	0.48	NA	NA	0.49	-0.07	0.49	-0.07
University education	1500	0.06	0.24	NA	NA	0.17	-0.10	0.17	-0.10
Previously employed	1571	0.38	0.49	NA	NA	0.47	-0.05	0.47	-0.05
Size of cohort	30	54.60	25.49	31.80	99.00	0.32	0.37	0.32	0.37
Program completion rate	30	0.86	0.13	0.73	1.00	0.53	-0.23	0.53	-0.23

4 Table 2: effects on LinkedIn account opening and usage

Table 2 is the first stage of any mechanism story: the training moved LinkedIn use, by a lot. Three outcomes: any account between baseline and 12 months, an account opened during training, and the usage index (first principal component of ten usage measures, standardized in the control group). The usage index rows also split the effect into an extensive margin (more accounts, valued at the control-group usage of account holders) and an intensive margin (more use per account). The decomposition needs only the account regression, the index regression, and the control mean of the index among account holders, so the shares reproduce from the public data even though the ten components do not.

```
t2 <- bind_rows(itt("li_account"), itt("account_in_bridge"), itt("summary_li_pca", cond = "li_account"))
ext <- t2$estimate[1] * t2$control_mean_cond[3]
shares <- c(extensive = ext / t2$estimate[3], intensive = (t2$estimate[3] - ext) / t2$estimate[3])
pub2 <- list(est = c(0.314, 0.395, 0.935), se = c(0.049, 0.048, 0.145), shares = c(0.319, 0.682),
             mean = c(0.484, 0.091, 0.000), cond = 0.950, n = c(1638, 1579, 1300), r2 = c(0.140, 0.260, 0.187))
tibble(
  ` ` = c("Treated cohort", "", "Share of treatment effect due to more accounts", "Share due to using accounts more",
         "Control group mean", "Control mean | account", "Number of respondents", "Number of cohorts", "Adjusted R2"),
  `(1) LinkedIn account` = c(fmt(t2$estimate[1]), paren(t2$std.error[1]), "", "", fmt(t2$control_mean[1]), "", fmt0(t2$n[1]), t2
  `(2) Account during training` = c(fmt(t2$estimate[2]), paren(t2$std.error[2]), "", "", fmt(t2$control_mean[2]), "", fmt0(t2$n[
  `(3) LI usage index` = c(fmt(t2$estimate[3]), paren(t2$std.error[3]), fmt(shares[1]), fmt(shares[2]), fmt(abs(t2$control_mean[
  `Published (1)` = c(fmt(pub2$est[1]), paren(pub2$se[1]), "", "", fmt(pub2$mean[1]), "", fmt0(pub2$n[1]), 30, fmt(pub2$r2[1])),
  `Published (3)` = c(fmt(pub2$est[3]), paren(pub2$se[3]), fmt(pub2$shares[1]), fmt(pub2$shares[2]), fmt(pub2$mean[3]), fmt(pub2
) %>%
  kbl(booktabs = TRUE, linesep = "", align = "lccccc",
      caption = "Table 2 of the paper: treatment effects on LinkedIn account opening and usage, block fixed effects, standard er
  tab_style("scale_down")
```

Table 2: Table 2 of the paper: treatment effects on LinkedIn account opening and usage, block fixed effects, standard errors clustered by cohort; columns 1 and 3 of the published table for comparison.

	(1) LinkedIn account	(2) Account during training	(3) LI usage index	Published (1)	Published (3)
Treated cohort	0.314	0.395	0.935	0.314	0.935
	(0.049)	(0.048)	(0.145)	(0.049)	(0.145)
Share of treatment effect due to more accounts			0.318		0.319
Share due to using accounts more			0.682		0.682
Control group mean	0.484	0.091	0.000	0.484	0.000
Control mean | account			0.950		0.950
Number of respondents	1,638	1,579	1,300	1,638	1,300
Number of cohorts	30	30	30	30	30
Adjusted R2	0.140	0.260	0.187	0.140	0.187

All three coefficients, their standard errors, the control means, and the two shares reproduce. The usage index effect of 0.935 standard deviations is the number the Appendix C shares divide by. About a third of it is more accounts and two thirds is more use per account, which the paper reads as candidates learning to use the platform rather than being pushed to open a dormant profile.

5 Figure 1: employment effects, with wild bootstrap p-values

Figure 1 is the paper's headline: employment rises by 7 percentage points at the end of the program and stays up at 6 and 12 months. The figure plots the control mean and the treated mean (control mean plus the fixed-effect ITT) at 0, 6, and 12 months with 95 percent intervals. With 30 clusters the paper adds wild cluster bootstrap p-values (Cameron, Gelbach, and Miller 2008), computed by the `wildboot` program in the replication package: Rademacher weights drawn once per cohort, 1,000 replications, the null imposed.

The wild bootstrap is defined here, in the note, not in the package, in the form the authors' program uses. Their `wildboot` program fits the unrestricted model with block fixed effects, forms the fitted values with the treatment coefficient set to zero and keeps the unrestricted residuals, then rebuilds the outcome in each replication by flipping the sign of each cohort's residuals with probability one half and refits. The p-value is the share of bootstrap t-statistics at least as large in absolute value as the observed clustered t. The textbook version (Cameron, Gelbach, and Miller 2008; Roodman et al. 2019) takes both the fitted values and the residuals from the restricted regression; the function offers both.

One detail of the shipped program matters for the published numbers. Inside `wildboot` the observed t-statistic uses the clustered standard error passed as an argument, but the bootstrap regressions reference a local macro named `myvce` that the program never defines, so each replication's t-statistic is computed with conventional standard errors. The `boot_vcov` argument reproduces that behaviour with `"iid"`; the clustered replications are `"cluster"`.

```
wild_cluster_boot_t <- function(data, y, d = "treated", fe = "strata", cluster = "studref",
                                n_boot = 1000, seed = 1, residuals = c("authors", "restricted"),
                                boot_vcov = c("cluster", "iid")) {
  residuals <- match.arg(residuals)
  boot_vcov <- match.arg(boot_vcov)
  dat <- data[complete.cases(data[, c(y, d, fe, cluster)]), ]
  cl_f <- as.formula(paste0("~", cluster))
  fit <- feols(as.formula(paste(y, "~", d, "|", fe)), data = dat, cluster = cl_f)
  b <- coef(fit)[[d]]
  t_obs <- coeftable(fit)[d, "t value"]
  if (residuals == "authors") {
    yhat <- fitted(fit) - b * dat[[d]]
    ehat <- resid(fit)
  } else {
    fit0 <- feols(as.formula(paste(y, "~ 1 |", fe)), data = dat)
    yhat <- fitted(fit0)
    ehat <- resid(fit0)
  }
  cl <- factor(dat[[cluster]])
  G <- nlevels(cl)
  vc <- if (boot_vcov == "cluster") cl_f else "iid"
  set.seed(seed)
  t_star <- vapply(seq_len(n_boot), function(r) {
    s <- sample(c(-1, 1), G, replace = TRUE)[as.integer(cl)]
    dat$.ystar <- yhat + s * ehat
    fb <- feols(as.formula(paste(".ystar ~", d, "|", fe)), data = dat, vcov = vc)
    coeftable(fb)[d, "t value"]
  }, numeric(1))
  list(estimate = b, std.error = coeftable(fit)[d, "Std. Error"], t = t_obs,
       p_wild = mean(abs(t_star) >= abs(t_obs)), p_cluster = coeftable(fit)[d, "Pr(>|t|)"],
       n_boot = n_boot, n_clusters = G)
}
```

```
emp <- c("employed1_all", "employed2_all", "employed3_all")
f1 <- bind_rows(lapply(seq_along(emp), function(k) {
  y <- emp[k]
  d <- wh[!is.na(wh[[y]]), ]
  b0 <- coef(feols(as.formula(paste(y, "~ treated")), data = d, cluster = ~studref))[["(Intercept)"]]
```

```
  r <- itt(y)
  wa <- wild_cluster_boot_t(wh, y, n_boot = n_boot_wild, seed = 1, residuals = "authors", boot_vcov = "cluster")
  wi <- wild_cluster_boot_t(wh, y, n_boot = n_boot_wild, seed = 1, residuals = "authors", boot_vcov = "iid")
  wr <- wild_cluster_boot_t(wh, y, n_boot = n_boot_wild, seed = 1, residuals = "restricted", boot_vcov = "cluster")
  tibble(month = c(0, 6, 12)[k], outcome = y, control = b0, estimate = r$estimate, std.error = r$std.error,
         treated = b0 + r$estimate, p_cluster = wa$p_cluster, p_wild_authors = wa$p_wild, p_wild_iid = wi$p_wild, p_wild_restric
}))
```

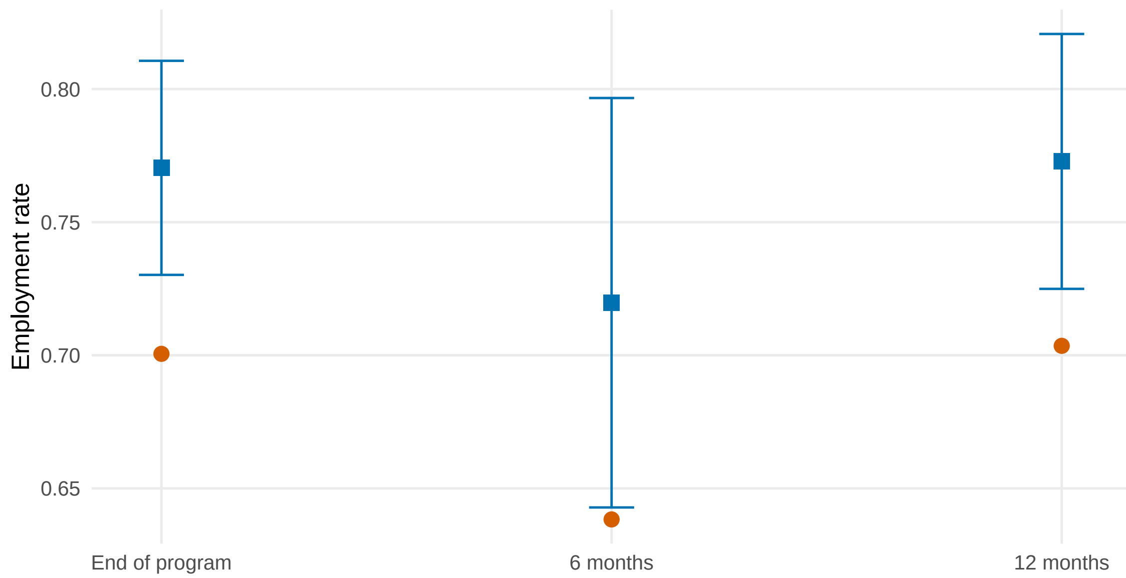

Figure 1: Figure 1 of the paper: employment status by survey wave. Red circles are control-group means; blue squares are the control mean plus the block-fixed-effect treatment effect, with 95 percent intervals from standard errors clustered by cohort.

Table 3: Employment effects behind Figure 1 with cluster-robust and wild cluster bootstrap p-values (1,000 Rademacher replications, 30 cohorts). 'Clustered' uses the authors' residual construction with clustered t-statistics in the replications; 'restricted' uses restricted residuals; 'as shipped' reproduces the program's conventional bootstrap t-statistics. The published column quotes the figure note: estimate (standard error) [wild bootstrap p-value].

Wave	Estimate	Std. error	Cluster p	Wild p, clustered	Wild p, restricted	Wild p, as shipped	Published
End of program	0.070	(0.021)	0.002	0.015	0.019	0.000	0.070 (0.021) [<0.001]
6 months	0.081	(0.039)	0.047	0.090	0.096	0.143	0.081 (0.039) [0.138]
12 months	0.069	(0.024)	0.008	0.016	0.037	0.000	0.069 (0.024) [<0.001]

The point estimates and standard errors reproduce, and the published p-values are reproduced only by the program as shipped. With clustered t-statistics in the replications, the wild bootstrap p-values are 0.015, 0.090, and 0.016; with the conventional standard errors the program actually uses they are 0.000, 0.143, and 0.000, which are the published "<0.001", 0.138, and "<0.001" up to Monte Carlo error. The mixed statistic is not a valid bootstrap-t (the observed and bootstrap t-statistics are on different scales), and in this design it understates the p-values at the end of the program and at 12 months. The conclusion does not change: the end-of-program and 12-month effects are significant by every version, and the 6-month effect is borderline by every version. The restricted-residual version gives 0.019, 0.096, and 0.037.

6 Table 3: effects on job type

Table 3 asks whether the new jobs are different jobs. Hours worked (winsorized), currently employed in the first job, more than one employer, permanent contract, and promotion, at 6 and 12 months. The outcomes are coded zero for the non-employed, so the conditional control means (among the employed) show whether the effect is more jobs or different jobs. The paper concludes it is more jobs: hours rise because employment rises, not because the employed work longer (Appendix B3 decomposes this).

```
jobtype <- c("hourswork_wins", "employ_1tcurrent", "numemployers_bin", "permemploy", "promoted")
labels3 <- c("Hours worked", "Employed in first job", "More than one employer", "Permanent contract", "Promoted")
t3a <- bind_rows(lapply(paste0("tr_", jobtype), itt, cond = "employed2_all"))
t3b <- bind_rows(lapply(paste0("tr_w2_", jobtype), itt, cond = "employed3_all"))
panel <- function(t, published) {
  tibble(` ` = c("Treated cohort", "", "Control group mean", "Control mean | employment", "Number of respondents", "Adjusted R2"
         !!!setNames(lapply(seq_len(nrow(t)), function(j) c(fmt(t$estimate[j]), paren(t$std.error[j]), fmt(t$control_mean[j]), f
}
pub3a <- list(est = c(4.200, 0.107, 0.001, 0.026, 0.007), se = c(1.700, 0.040, 0.021, 0.026, 0.010))
pub3b <- list(est = c(2.879, 0.126, -0.044, 0.034, -0.023), se = c(1.029, 0.026, 0.025, 0.025, 0.021))
bind_rows(panel(t3a, pub3a), panel(t3b, pub3b)) %>%
  kbl(booktabs = TRUE, linesep = "", caption = "Table 3 of the paper: treatment effects on job type six months (Panel A) and twe
  pack_rows("Panel A: six months after program completion", 1, 8) %>%
  pack_rows("Panel B: twelve months after program completion", 9, 16) %>%
  tab_style("scale_down")
```

Table 4: Table 3 of the paper: treatment effects on job type six months (Panel A) and twelve months (Panel B) after the program, with the published estimates in the last two rows of each panel.

	Hours worked	Employed in first job	More than one employer	Permanent contract	Promoted
Panel A: six months after program completion					
Treated cohort	4.200	0.107	0.001	0.026	0.007
	(1.700)	(0.040)	(0.021)	(0.026)	(0.010)
Control group mean	25.523	0.585	0.123	0.129	0.038
Control mean | employment	40.211	0.916	0.140	0.204	0.053
Number of respondents	1,107	1,117	1,114	1,113	1,117
Adjusted R2	0.078	0.077	0.007	0.104	0.001
Published estimate	4.200	0.107	0.001	0.026	0.007
Published SE	(1.700)	(0.040)	(0.021)	(0.026)	(0.010)
Panel B: twelve months after program completion					
Treated cohort	2.879	0.126	-0.044	0.034	-0.023
	(1.029)	(0.026)	(0.025)	(0.025)	(0.021)
Control group mean	29.233	0.602	0.144	0.189	0.118
Control mean | employment	41.590	0.855	0.148	0.269	0.155
Number of respondents	985	987	988	983	986
Adjusted R2	0.046	0.059	0.020	0.059	-0.001
Published estimate	2.879	0.126	-0.044	0.034	-0.023
Published SE	(1.029)	(0.026)	(0.025)	(0.025)	(0.021)

Every coefficient and standard error in both panels reproduces. Hours rise by about four per week at six months; the only job-quality outcome that moves is being in the first job at twelve months, which is retention rather than quality.

7 Table 4: the nonexperimental association

Table 4 is the first mediation-style piece of evidence, and the paper labels it nonexperimental. Among candidates, does having a LinkedIn account at baseline predict previous employment, and, in the control group, employment at the end of the program? The regressions are plain OLS with heteroskedasticity-robust standard errors; columns 2 and 4 add the covariates. In the language of Section 10, the column 4 coefficient is the delta of the Appendix C shares for a different mediator (the baseline account rather than the account by 12 months): a mediator-outcome association in the untreated, whose causal reading needs the mediator to be as good as random given the covariates.

```
covs_noemp <- setdiff(covs_miss, c("prev_employ_miss", "prev_employ_nomiss"))
ctrl <- wh[wh$treated == 0, ]
fits4 <- list(
  feols(prev_employ ~ s1_linkedin, data = wh, vcov = "hetero"),
  feols(as.formula(paste("prev_employ ~ s1_linkedin +", paste(covs_noemp, collapse = " + "))), data = wh, vcov = "hetero"),
  feols(employed1_all ~ s1_linkedin, data = ctrl, vcov = "hetero"),
  feols(as.formula(paste("employed1_all ~ s1_linkedin +", paste(covs_miss, collapse = " + "))), data = ctrl, vcov = "hetero"))
t4 <- sapply(fits4, function(f) c(coeftable(f)["s1_linkedin", 1:2], n = nobs(f), r2 = unname(r2(f, "ar2"))))
tibble(` ` = c("LI account at baseline", "", "Covariates included?", "Number of respondents", "Adjusted R2", "Published"),
      `(1) Previously employed` = c(fmt(t4[1, 1]), paren(t4[2, 1]), "", fmt0(t4[3, 1]), fmt(t4[4, 1]), "0.130 (0.035)"),
      `(2)` = c(fmt(t4[1, 2]), paren(t4[2, 2]), "Y", fmt0(t4[3, 2]), fmt(t4[4, 2]), "0.090 (0.035)"),
      `(3) Employed at end of program` = c(fmt(t4[1, 3]), paren(t4[2, 3]), "", fmt0(t4[3, 3]), fmt(t4[4, 3]), "0.068 (0.044)"),
      `(4)` = c(fmt(t4[1, 4]), paren(t4[2, 4]), "Y", fmt0(t4[3, 4]), fmt(t4[4, 4]), "0.035 (0.049)")) %>%
  kbl(booktabs = TRUE, linesep = "", caption = "Table 4 of the paper: nonexperimental relationship between employment and a Link
  tab_style("scale_down")
```

Table 5: Table 4 of the paper: nonexperimental relationship between employment and a LinkedIn account at baseline (columns 3-4 use control-group candidates only), heteroskedasticity-robust standard errors.

	(1) Previously employed	(2)	(3) Employed at end of program	(4)
LI account at baseline	0.130	0.090	0.068	0.035
	(0.035)	(0.035)	(0.044)	(0.049)
Covariates included?		Y		Y
Number of respondents	1,475	1,475	699	699
Adjusted R2	0.009	0.116	0.002	0.024
Published	0.130 (0.035)	0.090 (0.035)	0.068 (0.044)	0.035 (0.049)

The association is positive and shrinks by a third to a half once covariates enter, which is the paper's own warning about selection into LinkedIn use. The end-of-program association in the control group (column 4, 0.035) is not significant. The lecture's sensitivity analysis in Section 11 puts a number on how much residual selection would erase the mediated share.

8 Table 5: heterogeneity by baseline account

Table 5 asks whether the training helped those who already had an account, and finds the effect concentrated among those who did not. The specification interacts the treatment with the baseline account indicator and interacts the block fixed effects with it too, so each group has its own randomization strata. The published table reports the effect for candidates without an account (the main coefficient) and the difference for those with one. `reghdfe` drops singleton fixed-effect cells, which `fixest` does with `fixef.rm = "singleton"`; one cell is a singleton in each wave, hence the sample sizes.

```
wh$fe_times_li <- interaction(wh$s1_linkedin, wh$strata, drop = TRUE)
t5 <- bind_rows(lapply(emp, function(y) {
  d <- wh[!is.na(wh[[y]]) & !is.na(wh$s1_linkedin), ]
  fit <- feols(as.formula(paste(y, "~ treated + treated:s1_linkedin | fe_times_li")), data = d, cluster = ~studref, fixef.rm = "
  ct <- coeftable(fit)
  tibble(outcome = y, b_treat = ct["treated", 1], se_treat = ct["treated", 2], b_int = ct["treated:s1_linkedin", 1], se_int = ct
         mean_acc = mean(d[[y]][d$treated == 0 & d$s1_linkedin == 1]), mean_noacc = mean(d[[y]][d$treated == 0 & d$s1_linkedin =
         n = nobs(fit), adj_r2 = unname(r2(fit, "ar2")))
}))
pub5 <- list(b = c(0.083, 0.099, 0.087), sb = c(0.023, 0.041, 0.026), i = c(-0.031, -0.106, -0.067), si = c(0.030, 0.067, 0.051)
tibble(` ` = c("Treated cohort", "", "... x baseline LinkedIn account", "", "Control group mean | account", "Control group mean
       !!!setNames(lapply(1:3, function(j) c(fmt(t5$b_treat[j]), paren(t5$se_treat[j]), fmt(t5$b_int[j]), paren(t5$se_int[j]), f
                                              paste(fmt(pub5$b[j]), paren(pub5$sb[j])), paste(fmt(pub5$i[j]), paren(pub5$si[j])
  kbl(booktabs = TRUE, linesep = "", caption = "Table 5 of the paper: employment effects by prior LinkedIn account, block fixed
  tab_style()
```

The interaction is negative in every wave and never significant, which the paper reads as the training working through the account margin. In the language of Section 08, Table 5 is a group

Table 6: Table 5 of the paper: employment effects by prior LinkedIn account, block fixed effects interacted with the baseline-account indicator, standard errors clustered by cohort.

	(1) End of program	(2) 6 months	(3) 12 months
Treated cohort	0.083	0.099	0.087
	(0.023)	(0.041)	(0.026)
... x baseline LinkedIn account	-0.031	-0.106	-0.067
	(0.030)	(0.067)	(0.051)
Control group mean | account	0.777	0.740	0.833
Control group mean | no account	0.709	0.630	0.690
Number of respondents	1,518	1,073	949
Adjusted R2	0.053	0.080	0.062
Published: treated	0.083 (0.023)	0.099 (0.041)	0.087 (0.026)
Published: interaction	-0.031 (0.030)	-0.106 (0.067)	-0.067 (0.051)

average treatment effect by one baseline covariate. The package estimates the same two group effects from a doubly robust score with the blocks and covariates as controls, without a linear interaction model.

```
d5 <- wh[!is.na(wh$employed1_all) & !is.na(wh$s1_linkedin), ]
d5$account_baseline <- factor(ifelse(d5$s1_linkedin == 1, "account", "no account"))
sc5 <- dr_scores(d5, "employed1_all", "treated", x = c("strata", covs_miss), seed = 1)
g5 <- cate_gate(sc5, groups = "account_baseline", n_boot = 499, seed = 1)
g5$table
#>        group    n    share      estimate  std.error    conf.low conf.high
#> 1    account  250 0.164582 -0.0008542506 0.07553080 -0.14889190 0.1471834
#> 2 no account 1269 0.835418  0.0753085130 0.02646607  0.02343598 0.1271811
#>     band.low band.high
#> 1 -0.16635176 0.1646433
#> 2  0.01731802 0.1332990
```

Both approaches put the effect among candidates without an account and neither can resolve the effect among those with one. The score gives 0.075 for the 1,269 candidates without a baseline account, close to the regression's 0.083. For the 250 with an account it gives -0.001 against the regression's implied 0.052, a gap well inside the group's standard error of 0.076. That group is a sixth of the sample, so neither estimate says much about it; the regression borrows strength across groups through its functional form and the score does not.

9 Table 6: effects on potential mechanisms

Table 6 is the "effects on the mediators" step, and the paper runs it for ten candidate mechanisms. Supply-side information (jobs and profiles viewed), placement in the targeted sector, placement through a partner firm, demand-side information (profile completeness and views), number of connections, job applications on the platform, self-placement, program completion, an engagement index, and an aspirations index. Each is an ITT with block fixed effects. Four of the ten (the LinkedIn usage indices and platform applications) were built from data LinkedIn redacted; the six others reproduce.

```
mech <- c("targetsector", "harambee_partner_original", "selfplaced", "s2_completed", "summary_engage", "summary_aspire2")
labels6 <- c("(2) Employed in target sector", "(3) Employed at partner firm", "(7) Self-placed", "(8) Completed program", "(9) E
t6 <- bind_rows(lapply(mech, itt))
pub6 <- list(est = c(0.155, -0.024, -0.013, -0.023, 0.151, 0.065), se = c(0.034, 0.059, 0.009, 0.030, 0.122, 0.054),
             mean = c(0.539, 0.342, 0.056, 0.882, 0.000, 0.000), n = c(1626, 1626, 1626, 1612, 1246, 1231))
tibble(` ` = c("Treated cohort", "", "Control group mean", "Number of respondents", "Number of cohorts", "Adjusted R2", "Publish
       !!!setNames(lapply(seq_along(mech), function(j) c(fmt(t6$estimate[j]), paren(t6$std.error[j]), fmt(t6$control_mean[j]), f
  kbl(booktabs = TRUE, linesep = "", caption = "Table 6 of the paper, the six columns that the public data support: treatment ef
  tab_style("scale_down")
```

The six reproduced columns match the paper, and the pattern is the paper's argument for the mechanism. Treated candidates are 15 percentage points more likely to be placed in the targeted sector

Table 7: Table 6 of the paper, the six columns that the public data support: treatment effects on potential mechanisms with block fixed effects and cohort-clustered standard errors. Columns 1, 4, 5, and 6 of the published table (supply-side information 0.316 (0.073), demand-side information 0.842 (0.144), connections 0.559 (0.093), job applications 0.009 (0.004)) use redacted LinkedIn data.

	(2) Employed in target sector	(3) Employed at partner firm	(7) Self-placed	(8) Completed program	(9) Engagement index	(10) Aspirations index
Treated cohort	0.155	-0.024	-0.013	-0.023	0.151	0.065
	(0.034)	(0.059)	(0.009)	(0.030)	(0.122)	(0.054)
Control group mean	0.539	0.342	0.056	0.882	0.000	-0.000
Number of respondents	1,626	1,626	1,626	1,612	1,246	1,231
Number of cohorts	30	30	30	30	29	29
Adjusted R2	0.096	0.207	0.012	0.024	0.176	0.035
Published estimate	0.155	-0.024	-0.013	-0.023	0.151	0.065
Published SE	(0.034)	(0.059)	(0.009)	(0.030)	(0.122)	(0.054)
Published N	1,626	1,626	1,626	1,612	1,246	1,231

(finance and call centres), with no change in partner-firm placement, self-placement, completion, engagement, or aspirations. Together with the redacted columns (large effects on information and connections, none on on-platform applications), the paper concludes that LinkedIn worked as an information and signalling tool inside the program's own placement channel, not as a job board.

10 Appendix Table C1: the mediation shares

Table C1 converts the two first stages into a share of the employment effect explained by LinkedIn use, under two explicit assumptions. For a mediator M (the account by 12 months, or the usage index) the do-file estimates four regressions as a system with `suest` and cohort clustering: employment on treatment (β), M on treatment (γ), employment on treatment and M ($\tilde{\beta}$, α), and employment on M in the control group (δ). The shares are $S_1 = \alpha\gamma/\beta$ and $S_2 = \delta\gamma/\beta$ with delta-method standard errors from the system covariance. S_1 assumes that the mediator-outcome relation is unconfounded given treatment and block; S_2 replaces the mediator coefficient by its control-group value, which the paper motivates as "treatment increases LinkedIn use but does not change the relationship between LinkedIn use and employment".

`mediate_reg()` is this system. With `method = "delta"` it fits the same four regressions, computes the joint covariance from the stacked influence functions clustered by cohort (what `suest` does, with its $G/(G-1)$ cluster correction and no degrees-of-freedom factor), and reports the shares with their standard errors in `shares`. Block fixed effects enter as a factor covariate. Panel A's standard errors below are read from the same stacked covariance, so that they match `suest` rather than `reghdfe` (which adds the $(n-1)/(n-k)$ factor and gives 0.021 instead of 0.020 for β).

```
c1_fit <- function(m) {
  d <- wh[complete.cases(wh[, c("employed1_all", "treated", m)]), ]
  mediate_reg(d, "employed1_all", "treated", m, x = "strata", method = "delta", cluster = "studref")
}
fit_acc <- c1_fit("li_account")
fit_idx <- c1_fit("summary_li_pca")
c1_col <- function(f) {
  m <- f$spec$m
  c(fmt(coef(f$models$total)[["treated"]]), paren(sqrt(vcov_of(f, "total"))),
    fmt(coef(f$models$mediator[[1]])[["treated"]]), paren(sqrt(vcov_of(f, "mediator"))),
    fmt(coef(f$models$outcome)[["treated"]]), paren(sqrt(vcov_of(f, "outcome_d"))),
    fmt(coef(f$models$outcome)[[m]]), paren(sqrt(vcov_of(f, "outcome_m"))),
    fmt(coef(f$models$control)[[m]]), paren(sqrt(vcov_of(f, "control"))),
    fmt(f$shares$estimate[2]), paren(f$shares$std.error[2]), fmt(f$shares$estimate[3]), paren(f$shares$std.error[3]), fmt0(f$n))
}
vcov_of <- function(f, which) {
  # suest's cluster-robust covariance: G/(G-1) only, no (n-1)/(n-k) factor
  rows <- complete.cases(wh[, c("employed1_all", "treated", f$spec$m)])
  cl <- wh$studref[rows]
  n <- sum(rows)
  m <- f$spec$m
  se2 <- function(model, term, subset = seq_len(n)) {
    infl <- causalmetrics:::.cm_infl(model, n, subset)
```

```
    causalmetrics:::.cm_stacked_vcov(infl, cl)[term, term]
  }
  switch(which,
    total = se2(f$models$total, "treated"),
    mediator = se2(f$models$mediator[[1]], "treated"),
    outcome_d = se2(f$models$outcome, "treated"),
    outcome_m = se2(f$models$outcome, m),
    control = se2(f$models$control, m, which(wh$treated[rows] == 0)))
}
rows_c1 <- c("Treatment effect on employment (beta)", "", "Treatment effect on LinkedIn use (gamma)", "",
             "Treatment effect on employment | LinkedIn use (beta tilde)", "", "Employment-LinkedIn association | treatment (alp
             "Employment-LinkedIn association in control group (delta)", "", "S1 = alpha gamma / beta", "", "S2 = delta gamma /
tibble(` ` = rows_c1, `(1) LinkedIn account` = c1_col(fit_acc),
       `Published (1)` = c("0.070", "(0.020)", "0.321", "(0.049)", "0.019", "(0.026)", "0.158", "(0.027)", "0.146", "(0.026)", "
       `(2) LI usage index` = c1_col(fit_idx),
       `Published (2)` = c("0.083", "(0.020)", "0.954", "(0.145)", "0.028", "(0.025)", "0.058", "(0.014)", "0.059", "(0.017)", "
  kbl(booktabs = TRUE, linesep = "", caption = "Appendix Table C1 of the paper: relationship between treatment, initial employme
  pack_rows("Panel A: parameter estimates", 1, 10) %>%
  pack_rows("Panel B: share of treatment effect explained by LinkedIn use", 11, 14) %>%
  tab_style("scale_down")
```

Table 8: Appendix Table C1 of the paper: relationship between treatment, initial employment, and LinkedIn use. Panel A reports the system's parameters, Panel B the shares of the treatment effect explained by LinkedIn use with delta-method standard errors; block fixed effects in every regression, standard errors clustered by cohort.

	(1) LinkedIn account	Published (1)	(2) LI usage index	Published (2)
Panel A: parameter estimates				
Treatment effect on employment (beta)	0.070	0.070	0.083	0.083
	(0.020)	(0.020)	(0.020)	(0.020)
Treatment effect on LinkedIn use (gamma)	0.321	0.321	0.954	0.954
	(0.049)	(0.049)	(0.145)	(0.145)
Treatment effect on employment | LinkedIn use (beta tilde)	0.019	0.019	0.028	0.028
	(0.026)	(0.026)	(0.025)	(0.025)
Employment-LinkedIn association | treatment (alpha)	0.158	0.158	0.058	0.058
	(0.027)	(0.027)	(0.014)	(0.014)
Employment-LinkedIn association in control group (delta)	0.146	0.146	0.059	0.059
	(0.026)	(0.026)	(0.017)	(0.017)
Panel B: share of treatment effect explained by LinkedIn use				
S1 = alpha gamma / beta	0.729	0.729	0.668	0.668
	(0.306)	(0.306)	(0.253)	(0.253)
S2 = delta gamma / beta	0.672	0.672	0.678	0.678
	(0.266)	(0.266)	(0.259)	(0.259)
Sample size	1,626	1626	1,288	1288

Every entry, including the delta-method standard errors of the shares, reproduces to three decimals. The shares say that two thirds to three quarters of the employment effect would run through LinkedIn use if the mediator were unconfounded; their standard errors say that the interval covers everything from zero to more than the whole effect. The paper's phrase is "suggestive evidence".

11 Beyond the paper: natural effects, machine learning, sensitivity, and accounting

The four additions below keep the paper's data and specifications and ask what the Section 10 tools add. Each is labelled as ours.

11.1 Natural effects with an interaction

The shares are the linear model's proportion mediated; allowing the account's effect to differ by treatment changes little. With `interaction = TRUE` the outcome model adds treatment times mediator, and `mediate_reg()` reports both decompositions by simulation.

```
d_acc <- wh[complete.cases(wh[, c("employed1_all", "treated", "li_account")]), ]
fit_int <- mediate_reg(d_acc, "employed1_all", "treated", "li_account", x = "strata", interaction = TRUE,
                       cluster = "studref", n_sim = 1000, seed = 1)
fit_int$effects
#>        term    estimate  std.error    conf.low  conf.high
#> 1     total  0.06975830 0.02347837  0.02580847 0.11531887
#> 2       nde  0.01025584 0.03097423 -0.05073838 0.07050765
#> 3       nie  0.05950246 0.01924709  0.02374302 0.10018026
#> 4 nde_total  0.02485645 0.02579049 -0.02593424 0.07372338
#> 5  nie_pure  0.04490186 0.01124171  0.02496817 0.06859022
#> 6       cde -0.01158194 0.05056126 -0.11035066 0.08839641
fit_int$shares
#>                 share  estimate std.error
#> 1 proportion mediated 0.8529804  1.050213
```

The interaction coefficient is 0.045 and not significant: the account's association with employment is similar in both arms. The two decompositions differ by the mediated interaction, 0.015, and the proportion mediated is 0.85 with a simulation standard error of 1.05: a ratio whose denominator is a 3-standard-error total effect has a wide interval whichever model computes it.

11.2 Double machine learning with the covariates

The linear system conditions on blocks only; the influence-function estimator adds the ten covariates with flexible learners. The estimand is unchanged (sequential ignorability given block and covariates), so what changes is the credibility of the mediator model and the outcome model. Random forests for the outcome and logistic regressions for the two propensities; five folds.

```
rf <- lrn("regr.ranger", num.trees = 500, min.node.size = 20)
fit_dml <- mediate_dml(d_acc, "employed1_all", "treated", "li_account", x = c("strata", covs_miss),
                       learner_y = rf, folds = 5, n_rep = 3, seed = 1)
fit_dml$effects
#>        term    estimate  std.error     conf.low  conf.high
#> 1     total  0.05727157 0.02428874  0.009666525 0.10487662
#> 2       nde -0.00522391 0.02851392 -0.061110173 0.05066235
#> 3       nie  0.06249548 0.01488909  0.033313396 0.09167757
#> 4 nde_total  0.02606575 0.02781842 -0.028457344 0.08058884
#> 5  nie_pure  0.02913621 0.01558185 -0.001403649 0.05967608
```

The indirect effect with forests and covariates is 0.062 against 0.051 from the linear system. The covariates change the split little, which is what balance (Table 1) and the small covariate adjustment in Table 4 lead one to expect. The standard errors are of the same order; the influence-function estimator does not buy precision here, it buys robustness to the functional form.

11.3 Sensitivity to an unmeasured mediator-outcome confounder

How much residual correlation between account opening and employment would erase the share. `mediate_sensitivity()` traces the indirect effect against the correlation between the errors of the mediator and outcome regressions and reports where it crosses zero.

```
sens <- mediate_sensitivity(fit_acc, rho = seq(-0.5, 0.5, by = 0.02))
c(rho_zero = sens$rho_zero, rho_tilde = sens$rho_tilde)
#>  rho_zero rho_tilde
#> 0.1633502 0.1633512
plot_mediate_sensitivity(sens)
```

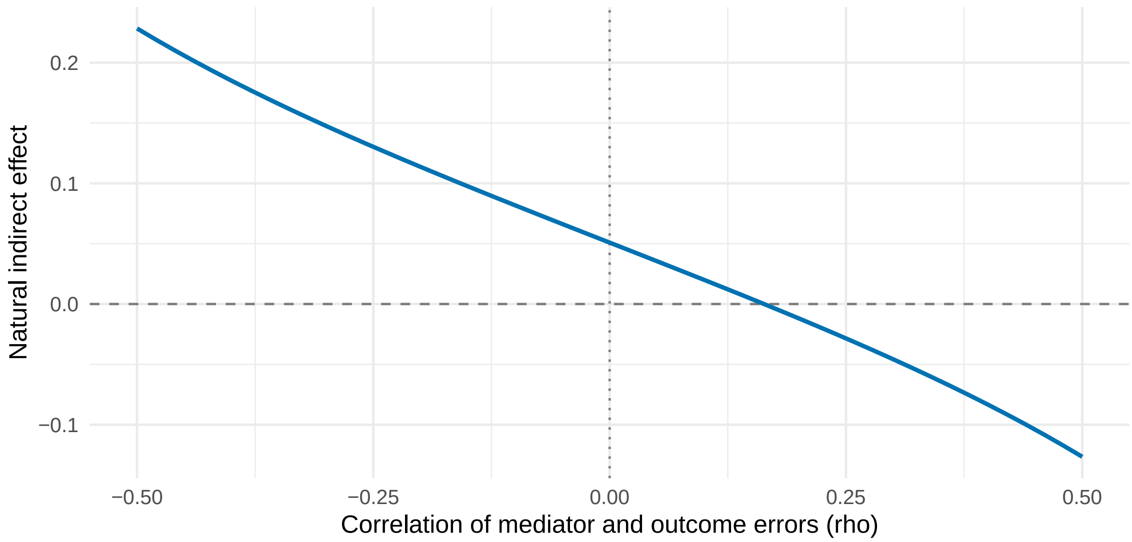

Figure 2: The indirect effect of the training through the LinkedIn account as a function of the correlation between the account and employment errors. Sequential ignorability is rho = 0; the effect vanishes at the value reported as rho zero.

An error correlation of 0.16 is enough. For scale, the covariates of Table 4 move the account coefficient from 0.130 to 0.090 among all candidates, a modest amount of observed selection; an unobserved trait of similar strength would take the share to zero. This is the quantitative form of the paper's caution.

11.4 Gelbach accounting of the mechanism variables

Adding the six public mechanism variables to the employment regression and attributing the change to each one. Gelbach's (2016) decomposition assigns the fall in the treatment coefficient to each added covariate without depending on the order of addition.

```
mech_g <- c("li_account", "targetsector", "harambee_partner_original", "selfplaced", "s2_completed", "summary_engage")
d_g <- wh[complete.cases(wh[, c("employed1_all", "treated", mech_g)]), ]
g_wh <- gelbach_decomp(d_g, "employed1_all", "treated", x_add = mech_g, fe = "strata", cluster = "studref",
                       n_boot = n_boot, seed = 1)
g_wh$coefficients
#>     term    estimate  std.error    conf.low   conf.high
#> 1   base  0.08712918 0.04394903  0.01392154 0.170506902
#> 2   full -0.04155684 0.03380527 -0.12261947 0.007128744
#> 3 change  0.12868601 0.04576452  0.04608729 0.218615462
g_wh$contributions[, c("covariate", "Gamma", "beta_full", "contribution", "std.error", "share")]
#>                  covariate        Gamma   beta_full  contribution   std.error
#> 4               li_account  0.333229767  0.02775508   0.009248818 0.005784236
#> 5             targetsector  0.191189859  0.62182588   0.118886802 0.039924155
#> 6 harambee_partner_original -0.078463419  0.03565020  -0.002797237 0.006943939
#> 7               selfplaced -0.008490273  0.55515789  -0.004713442 0.007767576
#> 8             s2_completed -0.019043612 -0.22067649   0.004202478 0.009790359
#> 9           summary_engage  0.152616509  0.02528295   0.003858596 0.009526360
#>         share
#> 4  0.07187120
#> 5  0.92385177
#> 6 -0.02173691
#> 7 -0.03662746
#> 8  0.03265683
#> 9  0.02998458
plot_gelbach(g_wh)
```

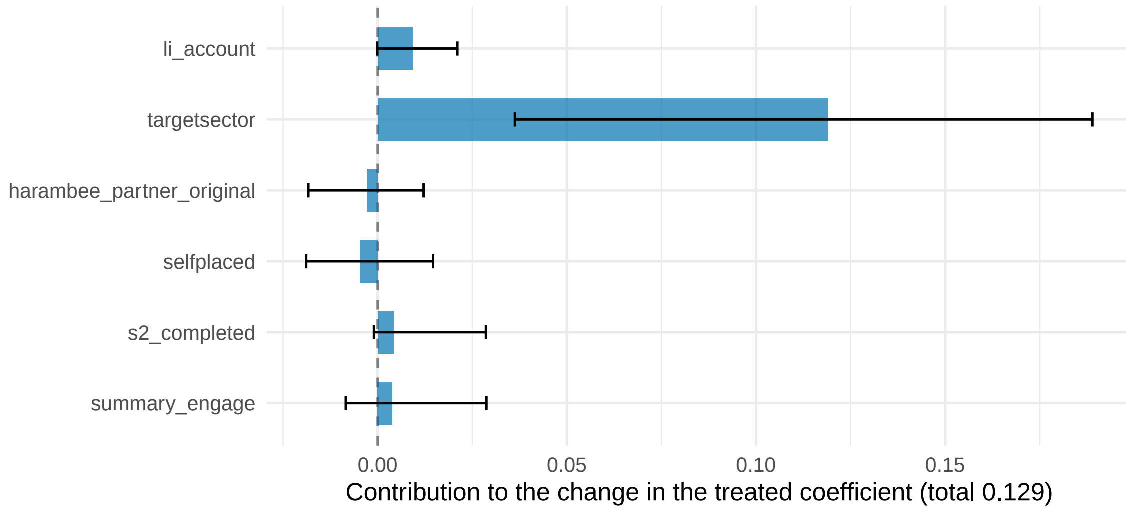

Figure 3: Gelbach decomposition of the end-of-program employment effect on the LinkedIn account and the five public mechanism variables, block fixed effects, cluster bootstrap intervals.

The decomposition is an accounting of where the jobs came from, not a mechanism. Placement in the targeted sector absorbs almost all of the coefficient because it is nearly the same event as being employed at the end of the program in this population. The LinkedIn account's own contribution is small once placement variables are in the regression, which is the lecture's point about post-treatment covariates that sit downstream of the mediator.

11.5 The first step by subgroup

Where did the training move the mediator? The account ITT by baseline account and by education, from the doubly robust score.

```
d_m <- wh[complete.cases(wh[, c("li_account", "treated", "s1_linkedin")]), ]
d_m$account_baseline <- factor(ifelse(d_m$s1_linkedin == 1, "account", "no account"))
d_m$education <- factor(ifelse(d_m$edn_dummy4 == 1, "university", ifelse(d_m$edn_dummy3 == 1, "post-secondary", "secondary")))
sc_m <- dr_scores(d_m, "li_account", "treated", x = c("strata", covs_miss), seed = 1)
cate_gate(sc_m, groups = "account_baseline", n_boot = 499, seed = 1)$table
#>        group    n     share  estimate  std.error    conf.low conf.high
#> 1    account  251 0.1651316 0.1132583 0.06315789 -0.01052885 0.2370455
#> 2 no account 1269 0.8348684 0.3670339 0.02745599  0.31322112 0.4208466
#>      band.low band.high
#> 1 -0.02821269 0.2547294
#> 2  0.30553360 0.4285341
cate_gate(sc_m, groups = "education", n_boot = 499, seed = 1)$table
#>            group   n      share  estimate  std.error     conf.low conf.high
#> 1 post-secondary 470 0.30921053 0.2956933 0.04439985  0.208671237 0.3827155
#> 2      secondary 957 0.62960526 0.3546536 0.03253829  0.290879737 0.4184275
#> 3     university  93 0.06118421 0.1700478 0.09068859 -0.007698571 0.3477942
#>      band.low band.high
#> 1  0.19367540 0.3977113
#> 2  0.27989008 0.4294171
#> 3 -0.03832821 0.3784238
```

12 What the replication says

The public data reproduce the paper. Tables 1 to 6, Figure 1, and Appendix Table C1 match to three decimals wherever the inputs are public; the four redacted Table 6 columns are the only gap. The one finding about the code is that the wild bootstrap program computes its replication t-statistics with conventional standard errors because of an undefined local macro; the clustered version gives larger p-values for the two significant waves without changing the conclusion.

The mediation content is the linear system of Section 10, run as a regression system with a delta method. `mediate_reg()` reproduces the shares and their standard errors because it computes the same stacked covariance. Its other outputs, the potential-outcome means and both decompositions, are free once the system is fitted.

The additions change no conclusion and sharpen the caveat. Forests and covariates leave the indirect effect where the linear system put it; an error correlation of about 0.16 would erase it; the Gelbach accounting shows that placement variables absorb the coefficient for a reason that has nothing to do with LinkedIn. The paper's own reading, "suggestive evidence" that the training worked through LinkedIn use, is the right one.

12.1 Mapping to the lecture's checklist

1. Estimand: total effect (RCT, Figure 1), effects on mediators (Tables 2 and 6), and a proportion mediated under sequential ignorability (Table C1). Stated as such in the paper.
2. Graph: treatment randomized; the account is a post-treatment choice whose confounders with employment (motivation, prior experience) are partly observed (Table 4).
3. Total and mediator effects first: done, with cluster-robust and wild bootstrap inference.
4. Natural effects with and without interaction, and with machine learning: this note's Section 11.
5. Sensitivity: rho at zero of 0.16.
6. No instrument for the mediator and no front-door structure.
7. No treatment-induced confounder is modelled; program completion is a candidate and moves little (Table 6).
8. Regression accounting: the Gelbach decomposition, labelled as accounting.

13 References

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